1 Lecture 11 Inference with OLS Short Answer Questions

2 Question #1

The least squares estimator is

$$b = (X'X)^{-1}X'Y.$$

Under the classical linear regression model assumptions, including normally distributed disturbances, the sampling distribution is

$$b \mid X \sim N\left(\beta, \sigma^2(X'X)^{-1}\right).$$

For confidence intervals and hypothesis testing for β_k, we use the standardized estimate. Explain how the standardized estimate is constructed and describe its sampling distribution.

3 Question #2

In the linear regression model:

$$Y = X\beta + \boldsymbol{\varepsilon},$$

define the residual maker for X as

$$M_X = I - X(X'X)^{-1}X'.$$

The OLS estimator b satisfies

$$Y = Xb + e.$$

a. Given that

$$e = M_X\boldsymbol{\varepsilon},$$

show that $e'e = \boldsymbol{\varepsilon}'M_X\boldsymbol{\varepsilon}$ (working through the matrix algebra)

b. Given that

$$\mathrm{E}(e'e \mid X) = \mathrm{tr}\left(M_X\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X)\right),$$

show that under homoskedasticity and non-autocorrelation,

$$\mathrm{E}\left(\frac{e'e}{n-K} \mid X\right) = \sigma^2.$$

(Please, no short cuts. Make all of your steps and simplifications clear)

4 Question #3

The least squares estimator is

$$b = (X'X)^{-1}X'Y,$$

where X is $n \times K$, Y is $n \times 1$, and b is $K \times 1$. Under the classical assumptions, the variance of the estimator is

$$\text{Var}(b \mid X) = \sigma^2 (X'X)^{-1}.$$

Consider the bivariate regression model

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i.$$

In this case, X is an $n \times 2$ matrix with a first column of ones and a second column containing the observed values of x_i:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \qquad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

a. Taking σ^2 as given, use matrix algebra to derive the variance of the OLS slope estimator b_2.

b. Using your result from Part (a), identify and explain the three factors that determine the variance of the OLS slope estimator.

5 Question #4

Consider the linear regression model:

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \cdots + \beta_K x_{iK} + \varepsilon_i.$$

a. Consider the OLS estimate b_3 of β_3 under the classical assumptions. Write down the formula for the variance of this estimator, taking σ^2 as given.

b. Explain what each of the three components of the formula represents and under what conditions they would improve the precision of the estimator.

6 Question #5

Consider the linear regression model:

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \cdots + \beta_K x_{iK} + \varepsilon_i.$$

a. Explain what multicollinearity is and how it affects the precision of OLS estimates.

b. Explain why removing highly correlated regressors may improve precision but can also introduce bias into OLS estimates.